WORKSHEET

Applications of binomial probability

1 A dart player typically hits the bullseye 70% of the time. After three shots, find:

a the expected number of times they hit the bullseye

b the standard deviation of the number of times they hit the bullseye

c the probability that they hit the bullseye all three times

d the probability that they miss the bullseye all three times

e the probability they hit the bullseye at least twice

f the probability that they do not hit the bullseye three times.

2 A line umpire has a 98% chance of making the correct decision on any call during a tennis match. If the umpire make 50 calls during the match:

a what is the probability that they make the correct decision on every call?

b what is the probability that they make one incorrect call?

c what is the probability that they make two incorrect calls?

d what is the probability that they make more than two incorrect calls?

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e what is the expected number of incorrect calls for the match?

f what is the standard deviation?

3 A test consists of 10 multiple-choice questions, each with five possible responses. If a student randomly guesses the answer to every question:

a use a CAS calculator or spreadsheet to find the probability of all possible outcomes

b create a graph for the probability distribution

c what is the probability that they get at least five correct?

4 A family has six children. Assuming males and females are equally likely, find:

a the expected number of males

b the probability of having no males

c the probability of having more males than females

d the probability of having one female.

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5 Two six-sided dice are rolled 60 times and the sum of the scores recorded. Find:

a the expected number of sevens rolled

b the standard deviation of the number of sevens rolled

c the probability that a score of seven occurs 10 times

d the probability that a score of seven occurs 20 times.

6 A manufacturer producing smartphone screens in high quantities typically finds that 2.5% of the screens are defective. A random sample of 70 is inspected for quality assurance. Find:

a the expected number of defects in the sample

b the probability that there are no defects in the sample

c the probability that there is one defect in the sample

d the probability that there are more than three defects in the sample

e the smallest number of screens that should be selected in a new sample to ensure a 90% chance of finding at least five defective screens.

7 A single card is randomly drawn from a standard deck of 52 cards before being replaced. How many times would the process have to be repeated to ensure a 99% chance or better of:

a drawing at least one ace?

b drawing more than 10 clubs?

c drawing more than 20 red cards?

d drawing more than three queens?

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8 Two cards are randomly dealt from a standard deck of 52 and then both are replaced. The process is repeated 10 times, and the number of pairs noted. Find:

a the expected number of pairs

b the probability of not getting a pair

c the probability of getting one pair

d the probability of getting three or more pairs.

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Answers

1 **a** 2.1

b ≈ 0.794

c 0.343

d 0.027

e 0.784

f 0.657

2 **a** ≈ 0.3642

b ≈ 0.3716

c ≈ 0.1858

d ≈ 0.0784

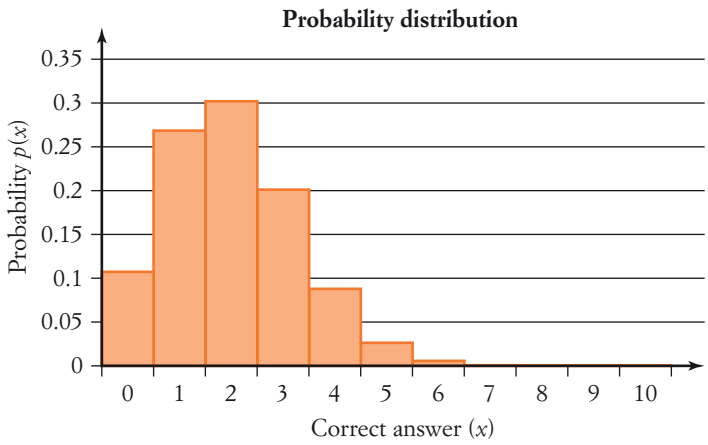
e 1

f ≈ 0.9899

3 **a**

x	$p(x)$
0	0.1074
1	0.2684
2	0.3020
3	0.2013
4	0.0881
5	0.0264
6	0.0055
7	7.864×10^{-4}
8	7.373×10^{-5}
9	4.096×10^{-6}
10	1.024×10^{-7}

b



c ≈ 0.0328

4 **a** 3

b ≈ 0.0156

c ≈ 0.3438

d ≈ 0.0938

5 **a** 10

b ≈ 2.887

c ≈ 0.1370

d ≈ 0.0008

6 **a** 1.75

b ≈ 0.1700

c ≈ 0.3050

d ≈ 0.0983

e 318

7 **a** 58

b 76

c 59

d 128

8 **a** ≈ 0.59

b ≈ 0.5454

c ≈ 0.3409

d ≈ 0.0179

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